

FINAL PROJECT REPORT

Streamlining IT Procurement: Automating Standard Laptop Orders with Flow Designer

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Team ID	SWTID-2026-9408
Project Name	Streamlining IT Procurement: Automating Standard Laptop Orders with Flow Designer

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1. Introduction

1.1 Project Overview

This project automates the procurement and configuration of standard laptops for employees using ServiceNow Flow Designer. When an employee's Standard Laptop Request is approved, the automation automatically creates and assigns a hardware configuration task to the Hardware team, tracks the request through to delivery, and keeps the requester informed at every stage — replacing a process that today depends on manual email/chat handoffs.

1.2 Purpose

IT procurement currently lacks an automated way to generate and assign hardware configuration tasks for standard laptop orders. This causes delays, manual overhead for the IT team, and inconsistent turnaround for employees — particularly noticeable during onboarding, when a late laptop directly delays a new hire's first productive day. The purpose of this project is to design and plan a ServiceNow Flow Designer solution that removes this manual handoff, so laptops are configured and delivered promptly with minimal manual intervention.

2. Ideation Phase

2.1 Problem Statement

Using the “I am / I'm trying to / but / because / which makes me feel” framework, three customer problem statements were defined for the personas affected by the current manual process:

PS	I am	I'm trying to	But	Because	Which makes me feel
PS-1	a new employee joining the company	get my laptop configured and ready so I can start work immediately	I often wait several days without any update	there's no automatic link between my approved request and a task for the Hardware team	frustrated and unsure who to follow up with
PS-2	a member of the IT Hardware support team	configure and dispatch every approved request as soon as possible	I usually only learn about an approved request days later	there's no automated task creation or assignment step	reactive and overloaded
PS-3	a member of the IT procurement team	move requests smoothly from approval to configuration	I have to manually track and follow up on every request	there's no workflow automation connecting the request to a task	stretched thin, worried requests will slip through

2.2 Empathy Map Canvas

To understand the requester's experience before the automation, an empathy map was built for the “New Employee Requesting a Standard Laptop” persona:

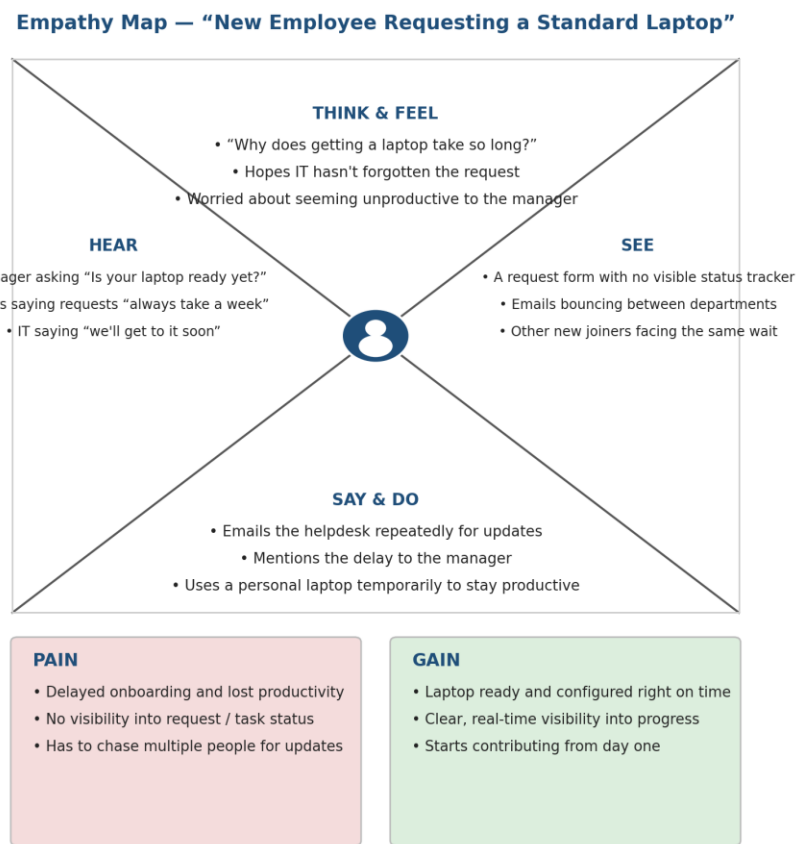


Figure 2.1: Empathy Map — New Employee Requesting a Standard Laptop

2.3 Brainstorming

The team brainstormed 15 ideas addressing the problem statement, grouped into four themes, then prioritized them by importance and feasibility:

- Cluster A — Flow Automation (Request → Task → Fulfilment): auto-create and auto-assign the hardware task, SLA escalation, CMDB auto-registration.
- Cluster B — Smart Approval Routing: cost-centre based routing, auto-approval under a threshold.
- Cluster C — Self-Service, Communication & Feedback: guided catalog form, Virtual Agent status checks, delivery notifications.
- Cluster D — Inventory & Vendor Readiness: buffer stock, vendor API integration, RPA purchase orders.

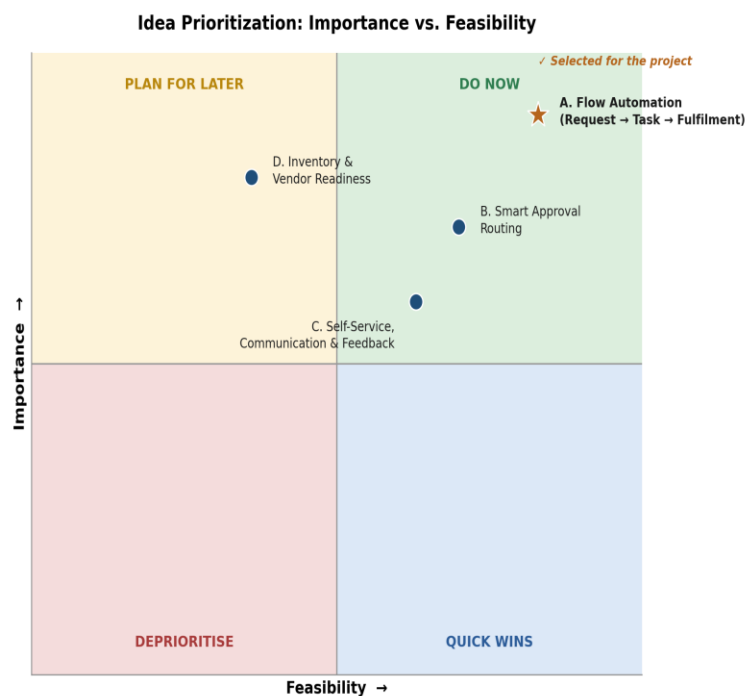


Figure 2.2: Idea Prioritization — Cluster A (Flow Automation) was selected for this project

Cluster A scored highest on both importance and feasibility, and directly matches the project's problem statement — it was selected as the idea to build via ServiceNow Flow Designer.

3. Requirement Analysis

3.1 Customer Journey Map

The current (as-is) journey for a standard laptop request was mapped across five phases — Entice, Enter, Engage, Exit, Extend — to surface friction points and map them to opportunities:

STREAMLINING IT PROCUREMENT

Customer Journey Map — Standard Laptop Request (Current / As-Is State)

SCENARIO: An employee (new hire or replacement case) needs a standard laptop, from the moment the need arises through approval, hardware configuration, delivery, and asset closure — before the Flow Designer automation described in this project.

	ENTICE		ENTER		ENGAGE			EXIT		EXTEND		
	How does someone become aware a laptop is needed?		How do they begin the process?		What happens in the core customer's?			What happens in the process lifecycle?		What happens after it's over?		
STEPS	New Hire Onboarding Trigger	Replacement Trigger	Browse Service Catalog	View Standard Laptop Item	Submit Request	Manager Approval	Manual Handoff to Hardware	Hardware Configures Laptop	Manual Status Update	Notification & Pickup	Asset Registration	No Structured Feedback Loop
INTERACTIONS	HR system, IT inbox, hiring manager	Manager, employee, informal chat / email	Employee Center portal, search bar	Catalog item page, generic request form	Catalog item, employee's own profile data	Manager, approval email / ServiceNow inbox	IT procurement staff, email/chat to Hardware team	Hardware team, laptop, imaging tools	ServiceNow record, Hardware team member	Employee, informal message or in-person handover	CMDB, IT asset manager	None formal, occasional verbal comments
GOALS & MOTIVATIONS	Help me make sure IT knows a laptop is needed before day one	Help me get a working laptop without a long delay	Help me find the right form quickly	Help me understand what I'm requesting	Help me submit this with minimal effort	Help me review and decide quickly	Help me make sure Hardware finds me promptly	Help me know exactly what to configure and for whom	Help me reflect that the laptop is ready	Help me know the moment it's ready to collect	Help me keep accurate inventory records	Help me share how the experience went
POSITIVE MOMENTS	Early heads-up gives IT some lead time	Manager is usually supportive of replacement	Catalog is easy to reach from the portal home page	A basic description of the laptop is available	The form itself is short and quick to fill	Manager can approve directly from email	A dedicated procurement person tracks handoffs	Hardware team is skilled and fast once they start	Status does eventually get updated	Employee is happy once they finally get the laptop	Assets do eventually get tracked	Employee do mention issues informally to their manager
NEGATIVE MOMENTS	No formal trigger — relies on someone remembering to email IT	No standard way to flag urgency or track the asset date	Search returns multiple, unclear laptop-related items	No visual guide comparing laptop tiers or specs	Department & manager fields must be typed in manually	Requester has no visibility into approval status	Delay if that person is busy, on leave, or forgets	Task details are often incomplete or scattered across emails	Update is a manual afterthought, often delayed	Employees often have to ask, or find out by chance	Registration is a separate manual step, sometimes skipped	No consistent data on satisfaction or turnaround time
AREAS OF OPPORTUNITY	How might Flow Designer improve this?											
	Auto-trigger a request from the HR onboarding record	A "Request Replacement" catalog option with reason capture	One clearly named "Standard Laptop Request" catalog item	Add a simple tier comparison (Basic / Developer / Power User)	Auto-fill department & manager from the user's profile	Show real-time approval status in "My Requests"	Flow Designer auto-creates & assigns the task the moment it's approved	Auto-generated task carries all request details & on SLA	Closing the task auto-updates the request stage to "Delivered"	Automatic email / notification the moment the task closes	Auto-register the asset in the CMDB when the task closes	A short automated survey triggered after delivery

Figure 3.1: Customer Journey Map — Standard Laptop Request (Current / As-Is State)

3.2 Solution Requirement

Functional Requirements:

FR No.	Epic	Sub Requirements
FR-1	Laptop Request Submission	Submit via Service Catalog; auto-fill department & manager; select laptop tier.
FR-2	Approval Workflow	Route to manager; auto-approve under a cost threshold; reject with reason.
FR-3	Automated Task Creation & Assignment	Auto-create hardware task on approval; auto-assign by location; set priority/SLA.
FR-4	Status Tracking & Notifications	Auto-update RITM stage; notify requester by email; IT dashboard of status.
FR-5	Asset & Inventory Update	Auto-register the laptop in CMDB; update stock count after dispatch.

Non-Functional Requirements:

NFR No.	Requirement	Description
NFR-1	Usability	Simple guided form with auto-filled fields; clear status visibility.
NFR-2	Security	Role-based access; approval respects org hierarchy; every action logged.
NFR-3	Reliability	Flow triggers reliably on every approval, with retry on failure.
NFR-4	Performance	Task creation and notification complete within seconds of approval.
NFR-5	Availability	Inherits ServiceNow's 99.8% uptime SLA; no manual keep-alive step.
NFR-6	Scalability	Reusable Flow Designer subflows handle rising request volume.

3.3 Data Flow Diagram

The Level-1 DFD below traces how request data moves through the system, from submission to asset registration:

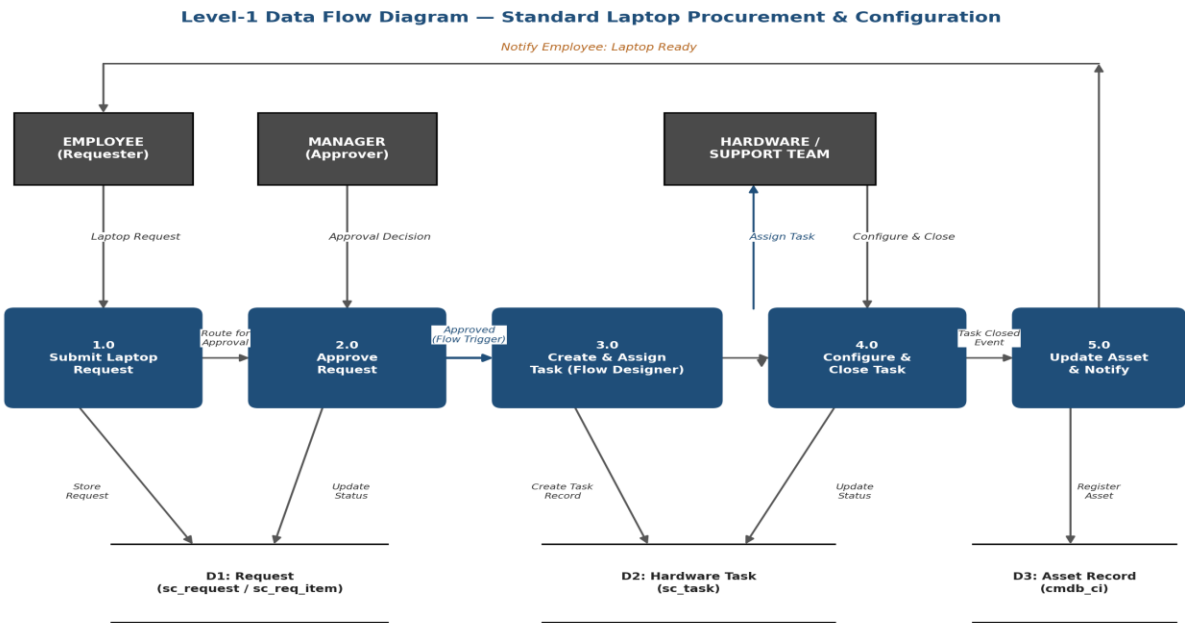


Figure 3.2: Level-1 Data Flow Diagram

3.4 Technology Stack

The solution is built entirely on native ServiceNow components, layered as follows:

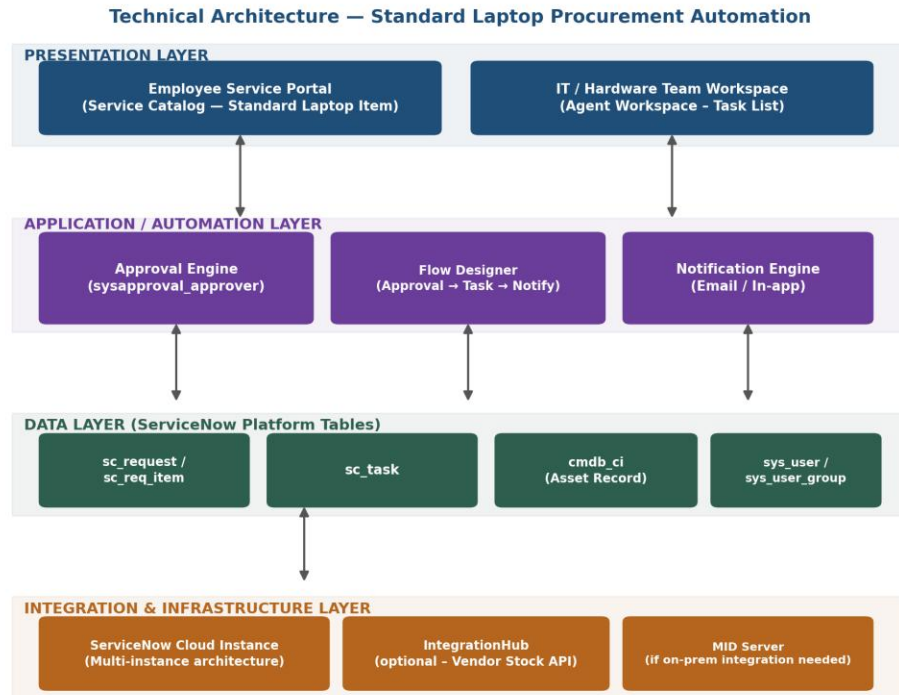


Figure 3.3: Layered Technical Architecture

4. Project Design

4.1 Problem – Solution Fit

The Problem-Solution Fit canvas confirmed genuine fit between the identified problem and the proposed automation:

- Customer Segment: Employees requesting a laptop, and the Hardware/Support team configuring it.
- Jobs-to-be-Done: Get a configured laptop to the employee quickly, without manual chasing.
- Root Cause: The catalog request has no automated link to a Hardware team task.
- Triggers: A new-hire record, a replacement request, or a stalled unassigned request.
- Your Solution: A Flow Designer flow that auto-creates and assigns the task on approval, updates status, and notifies the requester — removing the manual handoff entirely.

4.2 Proposed Solution

Problem Statement	IT procurement lacks an automated way to generate and assign hardware configuration tasks for standard laptop orders.
Idea / Solution	A Flow Designer automation that creates and assigns a hardware task on approval, tracks status, and notifies stakeholders.
Novelty	Uses native no-code Flow Designer — no custom scripting or external integration — with built-in SLA escalation and CMDB auto-registration.
Social Impact	Faster onboarding, less IT workload, higher employee satisfaction with IT services.
Business Model	Internal efficiency tool; value via cost-avoidance rather than revenue — fewer manual hours, faster turnaround.
Scalability	Reusable subflows extend the same pattern to other hardware/software requests and other departments.

4.3 Solution Architecture

The end-to-end component architecture and data flow are shown below:

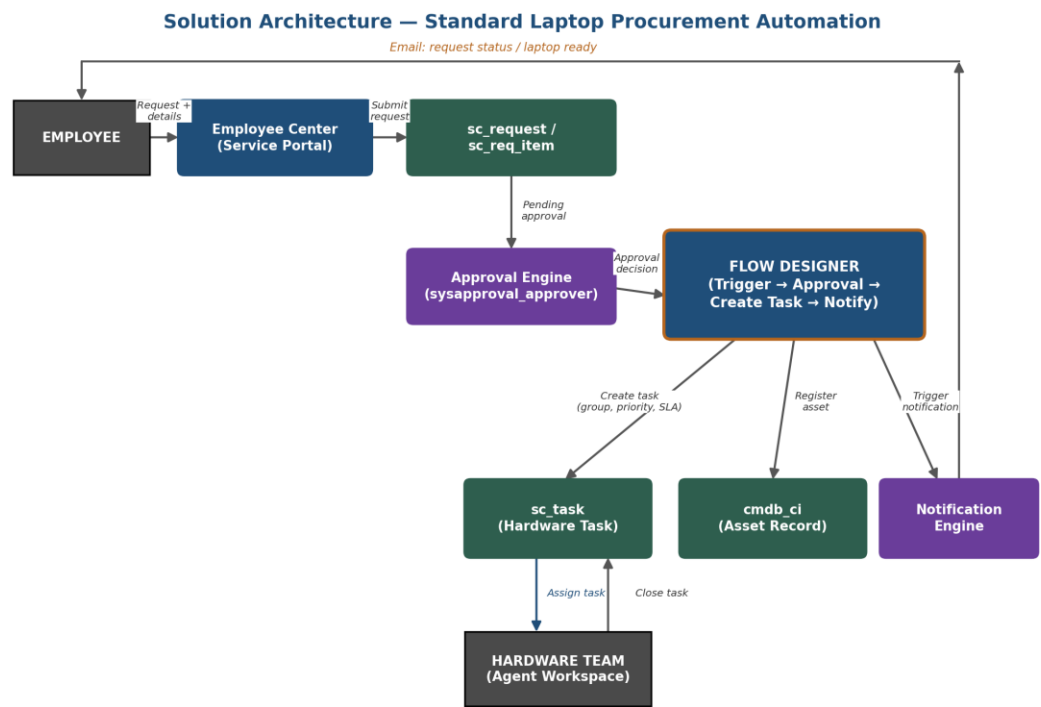


Figure 4.1: Solution Architecture — Standard Laptop Procurement Automation

5. Project Planning & Scheduling

5.1 Project Planning

The backlog of 10 user stories was estimated at 30 story points total and scheduled across two sprints:

Sprint	Focus (Epics)	Story Points	Duration	Planned Dates
Sprint-1	Laptop Request Submission, Approval Workflow, Automated Task Creation & Assignment	19	6 Days	28 Sep – 03 Oct 2026
Sprint-2	Status Tracking & Notifications, remaining Approval & Task items	11	6 Days	05 Oct – 10 Oct 2026

Average Velocity (Sprint-1) = 19 story points ÷ 6 days ≈ 3.2 points/day. The ideal burndown for Sprint-1:

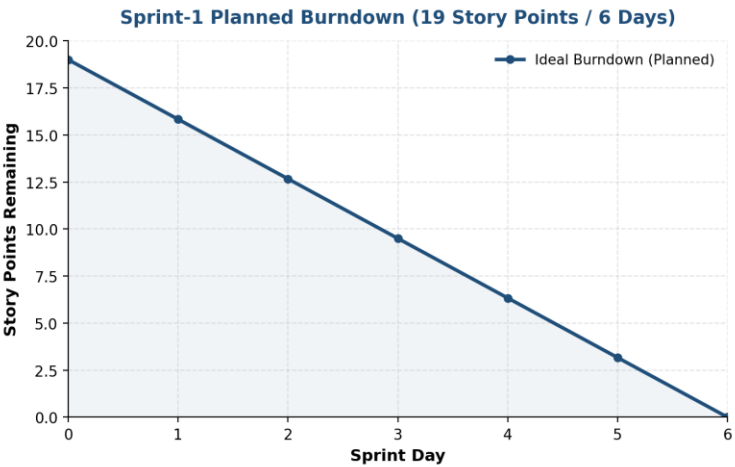


Figure 5.1: Sprint-1 Planned Burndown

6. Functional and Performance Testing

6.1 Performance Testing

User Acceptance Testing covers nine test cases spanning the full request lifecycle, including both positive and negative scenarios:

Test Case ID	Scenario	Type
TC-001	Submit a standard laptop request	Positive
TC-002	Manager approves the request	Positive
TC-003	Manager rejects the request	Negative
TC-004	Automated task creation & assignment	Positive
TC-005	Hardware team closes the task	Positive
TC-006	SLA escalation on delayed task	Positive
TC-007	Asset auto-registration in CMDB	Positive
TC-008	IT dashboard visibility	Positive
TC-009	Incomplete request submission	Negative

Full test steps, expected results, and the bug-tracking log are maintained in the separate User Acceptance Testing (UAT) document; results will be recorded there once execution begins.

7. Results

7.1 Output Screenshots

This project is at the design and planning stage; the Flow Designer flow, catalog item, and task automation described above have not yet been built in a live ServiceNow instance. Once implemented, this section will include screenshots of:

- The published “Standard Laptop Request” catalog item
- The Flow Designer canvas showing the trigger, approval, and task-creation actions
- A generated hardware task with auto-filled details
- The request record showing automatic stage updates

Workflow Studio New Flow Flow: None

Homepage Operations Integrations

Playbooks Flows Subflows Triggers Actions Decision tables

Flows 96 Last refreshed just now.

Name	Application	Status	Active	Updated
Application Intake Request Flow (Deprecated)	Application Intake	Published	false	2020-09-23 05:06
Application Intake Request V2	Application Intake	Published	true	2020-09-23 05:06
Benchmark Recommendation Evaluator	Benchmarks Spoke	Published	true	2026-04-30 01:01
Business process approval flow	Global	Published	true	2020-09-27 22:06
Change - Cloud Infrastructure - Authorize	Global	Published	true	2020-11-11 07:08
Change - Conflict Detection	Global	Published	true	2026-04-30 00:52
Change - Emergency - Authorize	Global	Published	true	2020-10-06 05:39
Change - Emergency - Implement	Global	Published	true	2020-09-23 05:06

New Playbook Flow Subflow Trigger Action Decision table Help me decide

Pick up where you left off

- Sub-Production tasks Last updated: 10 months ago by ...
- Production tasks Last updated: a year ago by Syste...
- Multi-factor Authentica... Last updated: a year ago by Syste...

Latest updates

- System Administrator modified Sub-Production tasks 10 months ago
- System Administrator modified Production tasks a year ago
- System Administrator modified Multi-factor Authentication a year ago

Workflow Studio New Flow Flow: None

Let's get the details for your flow

Flow name *

Application *

Description

Hide additional properties

Protection

Run as

Flow priority default

Cancel Build flow

Workflow Studio Standard laptop task Flow: Global

Standard laptop task Inactive

View: Test Activate

TRIGGER

Abort trigger creation

Trigger Select a Trigger

Q Search Triggers

- Record Record
- Scheduled Created
- Application Created or Updated Updated

ACTIONS Select multiple

Add an Action, F

ERROR HANDLER If an error occurs in your f

Data

Flow Variables

Trigger

No data available

Status: Draft Application: Global

Workflow Studio Standard laptop task Flow Global

Standard laptop task

Trigger: Service Catalog

Advanced Options

Delete Cancel Done

TRIGGER

Service Catalog

Flow Variables

- Trigger - Service Catalog
 - Requested Item Record Record
 - Run Start Time UTC Date/Time
 - Run Start Date/Time Date/Time
 - Table Name Table Name

ERROR HANDLING

If an error occurs

Status: Draft Application: Global

Workflow Studio Standard laptop task Flow Global

Standard laptop task

Action

Create Catalog Task

INSTALLED SPOKES

- ServiceNow Core
- ITSM

NOT INSTALLED SPOKES

- IntegrationHub

Service Catalog

Create Catalog Task

Flow Variables

- Trigger - Service Catalog
 - Requested Item Record Record
 - Run Start Time UTC Date/Time
 - Run Start Date/Time Date/Time
 - Table Name Table Name

ERROR HANDLING

If an error occurs

Status: Draft Application: Global

Workflow Studio Standard laptop task Flow Global

Standard laptop task

Action: Create Catalog Task

Action Inputs

Table Name: Catalog Task [sc_task]

* Requested Item [Requested Item]: Trigger -> Requested Item ...

Short Description: Laptop need to Configured

Description: Laptop need to Configured

Assignment group: Hardware

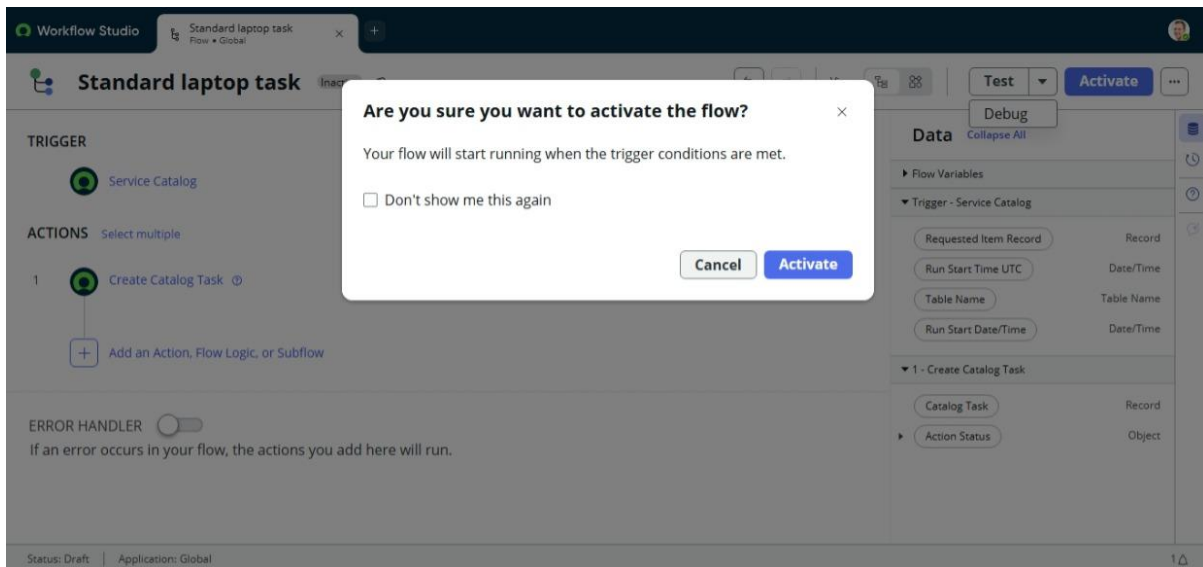
Approval: Approved

Wait: ☒

Flow Variables

- Trigger - Service Catalog
 - Requested Item Record Record
 - Run Start Time UTC Date/Time
 - Run Start Date/Time Date/Time
 - Table Name Table Name
- 1 - Create Catalog Task
 - Catalog Task Record
 - Action Status Object

Status: Modified Application: Global



Catalog Items									
Name	Short description	Active	Roles	Catalogs	Category	Price	Type	Updated	
Standard Laptop	Lenovo - Carbon x1	true		Service Catalog	Hardware	\$1,100.00	Item	2022-11-20 20:46:33	
StarTech Mini Display Port to DVI Adapter	PC / Mac Compatible	false		Service Catalog	Peripherals	\$17.23	Item	2022-11-20 20:46:33	
StarTech Mini Display Port to VGA Adapter	PC / Mac Compatible	true		Service Catalog	Peripherals	\$24.50	Item	2022-11-20 20:46:33	
StarTech USB Mini Hub	4 Port, USB 2.0	true		Service Catalog	Peripherals	\$11.55	Item	2022-11-20 20:46:33	
StarTech USB to DVI Adapter	Multi Monitor Adapter	true		Service Catalog	Peripherals	\$49.99	Item	2022-11-20 20:46:33	
STM MacBook Air Sleeve	STM Mac Sleeve - \$13"	true		Service Catalog	Peripherals	\$24.26	Item	2022-11-20 20:46:33	
STM MacBook Pro Sleeve	STM Mac Sleeve - \$15"	true		Service Catalog	Peripherals	\$25.78	Item	2022-11-20 20:46:33	
Table Index	Do you need to index a database table?	true		Service Catalog	Services	\$0.00	Item	2022-11-20 20:46:33	

Catalog Item - Standard Laptop

NameStandard Laptop

ApplicationGlobal

CatalogsService Catalog

CategoryHardware

State-- None --

Checked out-- None --

Owner

Secondary owners

Active☒

Fulfillment automation levelUnspecified

Item Details

Process Engine

Picture

Pricing

Portal Settings

Select the appropriate process engine for the catalog item. Only one engine can be selected.

FlowStandard laptop task

Workflow

Execution Plan

servicenow

AllFavoritesHistoryWorkspaces

Standard Laptop

Search

Service Catalog > Hardware > Standard Laptop

Search catalog

Lenovo - Carbon x1

x1 Carbon

The x1 Carbon is Lenovo's lightest ThinkPad yet. It provides a QHD display that fights glare and weighs less than three pounds. Ideal for most computing tasks, and highly mobile.

Technical Specs:

- Intel core i5 processor
- 512GB solid state drive (SSD)
- Backlit keyboard

Optional Software

- ☐ Adobe Acrobat
- ☐ Adobe Photoshop

Additional software requirements

Order this Item

Price

\$1,100.00 + \$100.00 Annually

Quantity

1

Subtotal

\$1,100.00 + \$100.00 Annually

Delivery time

0 Days

Order Now

Add to Cart

Shopping Cart

Empty

servicenow

AllFavoritesHistoryAdmin

Order Status: REQ0010001

Search

Order Status

Back to CatalogContinue ShoppingHome

Thank you, your request has been submitted

Order Placed: 2026-09-21 11:38:07

Request Number: REQ0010001

Estimated Delivery Date of Complete Order: 2026-09-21

Description (Includes Annual Charges)	Delivery Date	Stage	Price (ea.)	Quantity	Total
Lenovo - Carbon x1	2026-09-21		\$1,100.00 +\$100.00 Annually	1	\$1,100.00 +\$100.00 Annually
Total					\$1,100.00 +\$100.00 Annually

Back to CatalogContinue Shopping

Home

servicenow

AllFavoritesHistoryAdmin

Request - REQ0010001

Search

Request REQ0010001

DiscussFollowUpdateAdd New ItemCancel RequestCopyDelete

Activities: 1

System Administrator

Impact 3 - Low

Opened by System Administrator

Priority 4 - Low

State Open

Field changes • 2026-09-21 11:38:07

UpdateAdd New ItemCancel RequestCopyDelete

Requested Items (1)Approvers (1)Group approvals (1)Recurring Prices (1)

State Search

Actions on selected rows...Edit

Approval for = REQ0010001

State	Approver	Comments	Created
Requested	Eric Schroeder		2026-09-21 11:38:13

1 to 1 of 1

servicenow

AllFavoritesHistoryAdmin

Request - REQ0010001

Search

Request REQ0010001

DiscussFollowUpdateAdd New ItemCancel RequestCopyDelete

Activities: 1

System Administrator

Impact3 - Low

Opened bySystem Administrator

Priority4 - Low

StateOpen

Field changes • 2026-09-21 11:38:07

UpdateAdd New ItemCancel RequestCopyDelete

Requested Items (1)Approvals (1)Recurring Prices (1)

State

Copy URL to ClipboardCopy sys_id

Approval for = REQ0010001

Assign Tag

Approve

Reject

Requested

Approvers

Comments

Created

Eric Schroeder

2026-09-21 11:38:13

1 to 1 of 1

servicenow

AllFavoritesHistoryAdmin

Requested Item - RITM0010001

Search

Requested Item RITM0010001

DiscussFollowUpdateDelete

Activities: 1

System Administrator

Impact3 - Low

Opened bySystem Administrator

StateOpen

Field changes • 2026-09-21 11:38:07

UpdateDelete

Related Links

Flow Context

Catalog Tasks (1)Approvers

NumberSearch

Request Item = RITM0010001

Number

Assignment group

Assigned to

Short description

Actual start

Actual end

Order

SCTASK0010001

Hardware

(empty)

Laptop need to Configured

2026-09-21 11:41:14

(empty)

1 to 1 of 1

The screenshot shows a ServiceNow Catalog Task form for SCTASK0010001. The form is titled 'Catalog Task - SCTASK0010001' and includes a search bar. The form fields are organized into two columns. The left column contains: Number (SCTASK0010001), Assigned to (empty), Configuration item (empty), Active (checked), Short description (Laptop need to Configured), Description (Laptop need to Configured), and Work notes (empty). The right column contains: Approval (Approved), Priority (-- None --), State (Open), Request item (RITM0010001), and Requested for (System Administrator). Below the form fields are buttons for Discuss, Follow, Update, Close Task, and Delete. At the bottom, there is an Activities section showing a single activity for the system, with a field changes log indicating a change on 2026-09-21 11:41:14.

8. Advantages & Disadvantages

Advantages:

- Removes the manual handoff between approval and hardware configuration
- Faster, more consistent laptop turnaround for employees
- Automatic SLA escalation reduces missed or forgotten tasks
- Built with no-code Flow Designer — quick to build and maintain
- Automatic CMDB registration improves asset tracking accuracy
- Pattern is reusable for other standard hardware/software requests

Disadvantages / Limitations:

- Requires an active ServiceNow license and platform access
- Only covers standard, pre-defined laptop configurations — exceptions still need manual handling
- Depends on accurate, up-to-date catalog and CMDB data to work correctly
- Requires re-configuration if the approval hierarchy or Hardware team structure changes
- Initial build and testing effort before benefits are realized

9. Conclusion

This project designs an automated, ServiceNow Flow Designer–based solution to the inefficiency identified during the ideation phase: standard laptop requests currently rely on a manual handoff between IT procurement and the Hardware team, causing delays and inconsistent turnaround. Through empathy mapping, problem-solution fit analysis, and requirement and architecture design, the project defines a solution that automatically creates and assigns hardware configuration tasks on approval, tracks status, and notifies stakeholders — directly addressing the original problem statement and user story. The sprint plan and UAT test cases provide a clear, ready-to-execute path from design to a working, testable automation.

10. Future Scope

- Integrate with a vendor API (via IntegrationHub) for real-time stock levels and delivery ETA
- Extend the same automation pattern to other standard requests — monitors, software licenses, mobile devices
- Add a Virtual Agent / chatbot for self-service status checks
- Explore ServiceNow Predictive Intelligence to auto-categorize or auto-prioritize incoming requests
- Roll out across additional departments and office locations

11. Appendix

GitHub Link: <https://github.com/agnel-fernando-11/Streamlining-IT-Procurement-Automating-Standard-Laptop-Orders-with-Flow-Designer.git>

Demo Link: <https://drive.google.com/file/d/1H1nz85BTu1oQW1XO-SOTILedODUoYw2p/view?usp=sharing>